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2.1 Introduction

This chapter presents a review of the literature relevant to the development of our proposed system. It describes the core concept of enterprise architecture (EA) with the emphasis being put on its potential benefits and possible challenges. The discussion also delves into the different types of EA frameworks used to manage the architecture. A relevant example of it established in the higher education setting is also presented. The chapter further explores various sub-systems which make up the enterprise architecture of a university, such as student information, finance, learning and HR systems. Additionally, it takes a closer look at the current and past systems used at UTHM and their benefits. The chapter also presents the technology commonly used when designing such systems.

2.2 Enterprise Architecture (EA)

2.2.1 Definition and Concepts

Enterprise Architecture (EA) is a strategic procedure that helps an organisation manage complex operations and align their technologies, processes and resources with their business goals. Instead of allowing different systems and departments to operate separately, EA provides an overall blueprint showing how business functions, processes, information and technology platforms are interdependent (Kaisler et al., 2005). In today's rapidly changing business environment, organisations often struggle with isolated systems and unclear relationships between business and IT assets, which makes large-scale conversion work difficult and inefficient (Bloch et al., 2012). EA solves this problem by providing an integrated view of the enterprise and guiding the development of current and future architecture, so as to achieve more coordinated decision-making and more effective organisational changes (Niemi E., 2020).

There are two common EA frameworks are used to help organizations structure and manage their architecture:

1. Zachman Framework

Zachman Framework acts like a classification chart or a structured table for organizing all the important descriptions of an enterprise. It uses a matrix with six basic questions, “What, How, Where, Who, When, and Why”, and shows them from different viewpoints such as planners, designers, and builders. This ensures that the organization’s architecture is viewed from all angles and no important aspect is missed. (Zachman J., 2022)

2. TOGAF (The Open Group Architecture Framework)

TOGAF provides a step-by-step approach called the Architecture Development Method (ADM), which guides organisations through stages such as defining architectural vision, designing business and IT architecture, planning implementation and management of continuous improvement. This method helps to ensure that the architecture work is not only well-structured, but also practical and consistent with business goals. (David N., 2025)

2.2.2 Review of Previous Studies (University Enterprise Architecture)

Enterprise Architecture (EA) has established itself as a crucial tool in higher education to integrate all components of the organisation from IT systems to academic programs in such a way that they are effectively aligned with the objectives of the institution. In order to improve the organisation of its information systems, for example, the Universitas Nahdlatul Ulama Surabaya in Indonesia implemented the TOGAF framework. This change accelerated the workflows at the university and improved the exchange of data between departments. Another example is a university in Europe whose use of EA to drive digital transformation has strengthened the institution's ability to adapt to new trends in education and service delivery.

Still, many universities face challenges in implementing Enterprise Architecture (EA). Changes sometimes encounter many problems or it can be difficult to get everyone in the institution to get involved on board. In addition, these frameworks, often developed for the business sector, may appear rigid or inappropriate in academic cultures. However, as universities adapt to their own context and receive the support of a strong management level, the benefits become clear for better system integration, more flexible processes and progress in achieving their strategic goals.

2.3 Related Sub-Systems

This section highlights the subsystems from UTHM's core functions such as finance and human resources. Each subsystem plays a crucial role in the UTHM's educational system.

2.3.1 Student Information System (SIS)

A Student Information System (SIS) is a software platform used by educational institutions to store, manage and process all student-related data during their time as a student. Academic information systems are one of the efforts of educational institutions to provide quality services to provide convenience and comfort to students (Saputuri & Mulyani, 2022). Because of the role it plays, this system is a crucial module within the larger university ERP system.

At Universiti Tun Hussein Onn Malaysia (UTHM) it is the Student Academic Information System (SMAP Online) which serves this purpose. It is made up of six modules: Bulletin, Biodata, Registration, Examination, Graduation and Student's Account. The system encompasses key functionalities such as course registration, exam results, academic announcements, admissions and others. Prior to the introduction of this system the different functions were handled across different university systems. The introduction of the centralized SMAP platform was a huge improvement to the outdated solutions, as it allows all the student-related information to be found in one place and be easily accessible by students and staff anywhere, anytime. As noted by Musa (2021), SMAP is a key pillar of UTHM's smart academic management system.

2.3.2 Financial System

This system is represented by UTHM's ePayment and other financial modules that manage all the financial transactions in UTHM. This includes student fees, employee salaries, procurement and financial reporting. It directly maps to the "finance" module of a comprehensive enterprise resource planning (ERP) system.

Previously, financial systems such as ePayment, payroll and procurement modules were usually built on **monolithic or tightly-coupled legacy architectures**. These old systems can effectively handle core financial tasks, but they are difficult to update, the integration with new services is slow, and the maintenance cost is high. Because each function, such as fee collection, payroll processing or procurement, is developed as a separate component

and interoperability is limited, the system lacks flexibility and real-time data sharing. This reflects the challenges described in the legacy financial system literature. This outdated architecture limits the scalability and needs modernisation to support integrated ERP-style financial management. (Sreenivasulu G., 2025)

2.3.3 Human Resource (HR) System

The Human Resources (SDM) system previously used by the University of Tun Hussein Onn Malaysia (UTHM) runs on an outdated independent platform. Although the old system can perform some basic human resources functions, such as recording attendance and storing employee information, most of its processes are isolated from each other. Therefore, tasks such as updating employee information or preparing salary statements require manual data transmission and repeated input. This not only reduces work efficiency but also increases the probability of mistakes.

In view of these limitations, UTHM decided to upgrade to a more modern and comprehensive integrated human resources management system. This upgrade has contributed to the launch of the "SMART HRM" platform, which integrates daily human resources processes into an interconnected environment. The new system can be connected to other university databases, support remote access by employees and managers, and provide functions such as online leave application, self-service personal information update portal and real-time performance monitoring.

Through these improvements, the human resources activities of various departments can share information more easily, operate more efficiently, and avoid delays caused by old-style monomer systems. The adoption of SMART HRM is also in line with the global trend of human resources digitalisation, ensuring that UTHM's human resources management can support the long-term strategic needs of universities.

2.3.4 E-Learning System

University Tun Hussein Onn Malaysia (UTHM) has an online learning system to support online and hybrid learning initiatives of the university. This online learning system is also known as Author UTHM, and it is mainly used for course delivery, evaluation, and as a medium to connect students and lecturers.

Previously, UTHM's online education relied on many different platforms, and teachers shared information through social network groups, Google Classroom, email, and other media. Due to the lack of integration of these tools with the university's main databases, there are problems such as inaccurate records, difficulties with assessment management, and challenges in monitoring student participation. As digital learning has been normalised in higher education, therefore, an integrated and structured system is needed for UTHM.

Author UTHM has overcome these limitations by creating a unified learning management system (LMS) that integrates functions such as course content, homework submission, tests, and teacher-student interaction. The platform supports automatic marking, real-time updates, and class materials, including notes and videos. The integrated login system allows users to access resources and submit homework easily, and also reduces the administrative workload of faculty and staff.

2.4 Technology Used

In order to support daily operations, the enterprise systems used by universities rely on a variety of software technologies, architectures and infrastructure components. These technologies ensure that each subsystem is efficient, safe, scalable, and can be integrated with other university functions.

2.4.1 Relational Database Management Systems

Relational databases, such as MySQL, MariaDB, or Oracle database, are crucial for almost all university enterprise systems. These databases store structured data, such as student files and academic records, grades, user activity logs, financial transactions, payment logs, attendance logs, and staff records. Since this technology can ensure data consistency, reliability, fast retrieval and secure access, it is very suitable for large-scale institutional applications.

2.4.2 Web-Based Application Frameworks

Most subsystems of UTHM, such as SIS, financial system, HR system and online learning system, are developed using Web-based application frameworks. For example, the development of the front-end interface uses HTML5, CSS3 and JavaScript, while the back-end processing uses PHP, Java or .NET. These frameworks enable staff and students

to access the system through a single Web browser, which led to platform-independent and quick access. In addition, Web-based development supports responsive design, so that the system can be used on PCs, tablets and mobile devices.

2.4.3 Cloud Hosting, Backup, and High Availability Infrastructure

UTHM's systems have always relied on a server infrastructure that includes cloud or virtualized hosting environments, load balancing and failover support, automated backup systems, and high availability settings. The infrastructure minimizes downtime, prevents data loss, and ensures continuous access to the system. High-availability hosting is especially important for key subsystems, especially during peak hours. The example of peak hours for subsystems are the financial subsystem during the payroll period, the Student Information System (SIS) during the registration week, and during the online tests and quizzes. This infrastructure helps to avoid system crashes due to excessive traffic.

2.4.4 Integration Technologies and Middleware

UTHM uses technologies such as middleware platforms, SOAP Web services and RESTful APIs to facilitate communication between subsystems. These support real-time data sharing between multiple subsystems, including student information system (SIS), human resources (HR), finance and online learning systems. For example, the employee information in the human resources system can be connected to the salary module in the financial system, and the student registration data in the student information system can be immediately synchronized with the online learning system. Integrated technology helps the department to smooth the workflow and avoid duplication of work.

2.5 Summary

The reviewed literature shows that by adopting Enterprise Architecture, institutions can significantly simplify and accelerate their day-to-day operations by organizing their workloads and integrating technology, processes and resources into a single unified system. The use of EA frameworks shows to further help organizations to effectively manage the architecture in practice. Additionally, the real-life example of EA being successfully implemented in a university in Indonesia illustrates the possibilities of successful system adaptation in the higher education sector. In this case, EA not only improves process

efficiency but also allows systems to be flexible and customizable to specific institutional needs. Moreover, research on UTHM operations highlights the importance of up-to-date and practical sub-systems as they play a crucial part in effective and accessible institutional systems.

Reference

Bloch M, Blumberg S, Laartz J (2012) Delivering large-scale IT projects on time, on budget, and on value.

<http://www.mckinsey.com/business-functions/business-technology/our-insights/delivering-large-scale-it-projects-on-time-on-budget-and-on-value>

Kaisler SH, Armour F, Valivullah M (2005) Enterprise architecting: critical problems.

<https://doi.org/10.1109/HICSS.2005.241>

Niemi E, Pekkola S (2020) The Benefits of Enterprise Architecture in Organizational Transformation. <https://doi.org/10.1007/s12599-019-00605-3>

Zachman J (2022) The Framework for Enterprise Architecture: Background Description and Utility.

<https://zachman-feac.com/the-framework-for-enterprise-architecture-background-description-and-utility>

David N (2025) TOGAF ADM Explained: A Guide to the Architecture Development Method.

<https://www.itar.pro/togaf-adm-explained>

Sreenivasulu G. (2025). Architectural Transformation of Legacy Financial Systems: A Framework for Microservices, Cloud, and API Integration.

https://doi.org/10.34218/IJTMIS_16_02_075

Ajer, A. K. S., & Phan, T. (2023). Institutional work for enterprise architecture. *ECIS 2023 Research Papers*, 132. AIS Electronic Library (AISeL).

https://aisel.aisnet.org/ecis2023_rp/132

Herlambang, T., Siswantoro, J., Suryadi, K., & Handayani, P. W. (2021). Enterprise architecture framework in higher education. *Jurnal Sistem Informasi* (Journal of Information Systems), Universitas Nahdlatul Ulama Surabaya.

<https://journal2.unusa.ac.id/index.php/JSI/article/view/1611>

Rahmat, M., Shamsuddin, S. M., Sani, N. F. M., Komariah, K., & Hanapi, Z. (2022). An enterprise architecture planning for higher education using the TOGAF framework. *IEEE Xplore Digital Library*.

<https://ieeexplore.ieee.org/document/9947946>

Awadid, A., & Romdhane, R. (2020). Enterprise architecture adoption challenges. *Procedia Computer Science*, 176, 1415–1424. ScienceDirect.

<https://www.sciencedirect.com/science/article/pii/S1877050920314010>

Sodiq, S. S., Sarno, R., & Wicaksono, I. (2022). The role of enterprise architecture in enhancing digital transformation in higher education. *International Journal of Engineering Research & Development (IJERD)*. <https://www.ijerd.com/paper/vol18-issue1/I18014350.pdf>

Hussein, L. K., & Abdullah, N. (2023). Technology Readiness of Smart Human Resource Management System at Universiti Tun Hussein Onn Malaysia. *Research in Management of Technology and Business*, 4(1), 526-537.

<https://publisher.uthm.edu.my/periodicals/index.php/rmtb/article/view/1043>

Saputri, A., & Mulyani, E. (2022). *The impact of academic information system to students' satisfaction in higher education*. *Studies in Educational Management*, 12(1), 1–9.

<https://doi.org/10.32038/sem.2022.12.01>

Musa, M., Ismail, M. N., & Md Fudzee, M. F. (2021). *Smart campus implementation in Universiti Tun Hussein Onn Malaysia: Towards a conceptual framework*. *Journal of Physics: Conference Series*, 1860(1), Article 012008.

<https://doi.org/10.1088/1742-6596/1860/1/012008>